

Microprocessor Systems

Lab8 - External Interrupt & PWM

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8051 Interrupts

- Interrupt has external or internal event that interrupts the microcontroller to inform that a device needs its service
- In 8051 microcontroller there are 2 external interrupts, 2 timer interrupts, and 1 serial interrupt
 - External interrupts: External interrupt 0 (INT0) and external interrupt 1 (INT1)
 - Timer interrupts: Timer 0 interrupt and timer 1 interrupt
 - Serial interrupt is given for serial communication with the microcontroller

Vector Address

Interrupt	Flag	Interrupt vector address
Reset	-	0x0000
External interrupt 0 (INT0)	IE0	0x0003
Timer 0	TF0	0x000B
External interrupt 1 (INT1)	IE1	0x0013
Timer 1	TF1	0x001B
Serial	TI/RI	0x0023

Interrupt Priority

Interrupt	Flag	Interrupt priority
External interrupt 0 (INT0)	IE0	0 (High)
Timer 0	TF0	1
External interrupt 1 (INT1)	IE1	2
Timer 1	TF1	3
Serial	TI/RI	4 (Low)

Interrupt Enable (IE) Register

- All interrupt are disabled after reset
- IE register is used to enable/disable interrupt sources

EA	-	-	ES	ET1	EX1	ET0	EX0
----	---	---	----	-----	-----	-----	-----

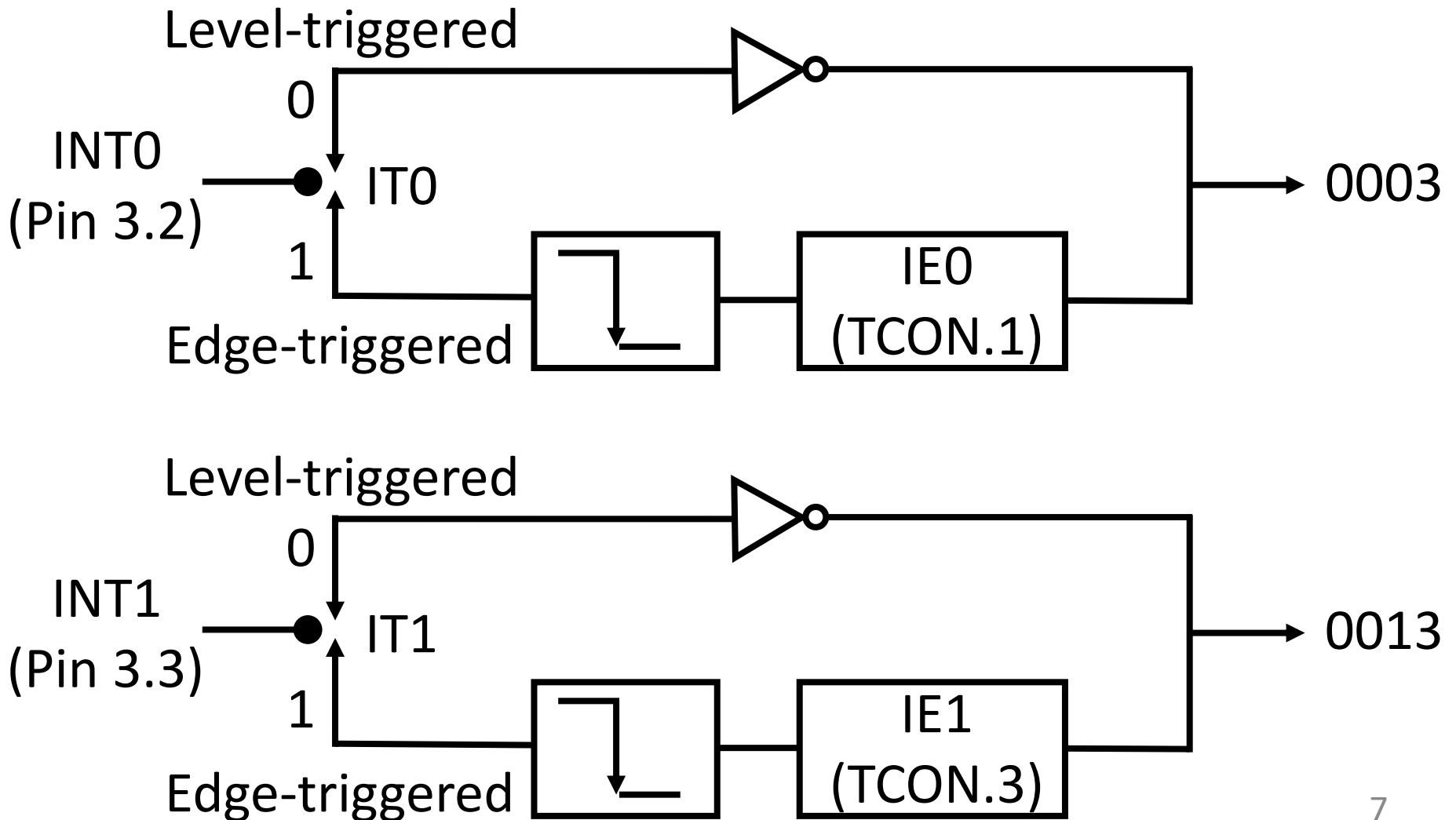
EA	Enables/disables all interrupts
-	Reserved for future use
-	Reserved for future use
ES	Enables/disables serial port interrupt
ET1	Enables/disables timer1 overflow interrupt
EX1	Enables/disables external interrupt1 (INT1)
ET0	Enables/disables timer0 overflow interrupt
EX0	Enables/disables external interrupt0 (INT0)

Time Control Register (TCON)

Bit No.	7	6	5	4	3	2	1	0
Symbol	TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

- IT0 (IT1): Interrupt 0 (1) type control bit
 - IT = 1 -> Edge-triggered
 - IT = 0 -> Level-triggered
- IE0 (IE1): External interrupt 0 (1) edge flag
 - IE = 1 when external interrupt edge is detected
 - IE = 0 when interrupt is finished

External Interrupts in 8051



Example (External Interrupt)

```
unsigned char LED=0x80;
```

```
main()
```

```
{
```

```
EA=1;
```

```
EX0=1;
```

```
EX1=1;
```

```
TCON=0x01;
```

```
while(1)
```

```
{
```

```
    P1=~LED;
```

```
    delay_ms(1000);
```

```
    LED>>=1;
```

```
    if(LED==0)
```

```
        LED=0x80;
```

```
}
```

```
}
```



- EA=1: Enable interrupts
- EX0=1: Enable external interrupt 0 (INT0)
- EX1=1: Enable external interrupt 1 (INT1)
- IT0=1: Edge-triggered
- IT1=0: Level-triggered

Example (External Interrupt)

- Edge-triggered

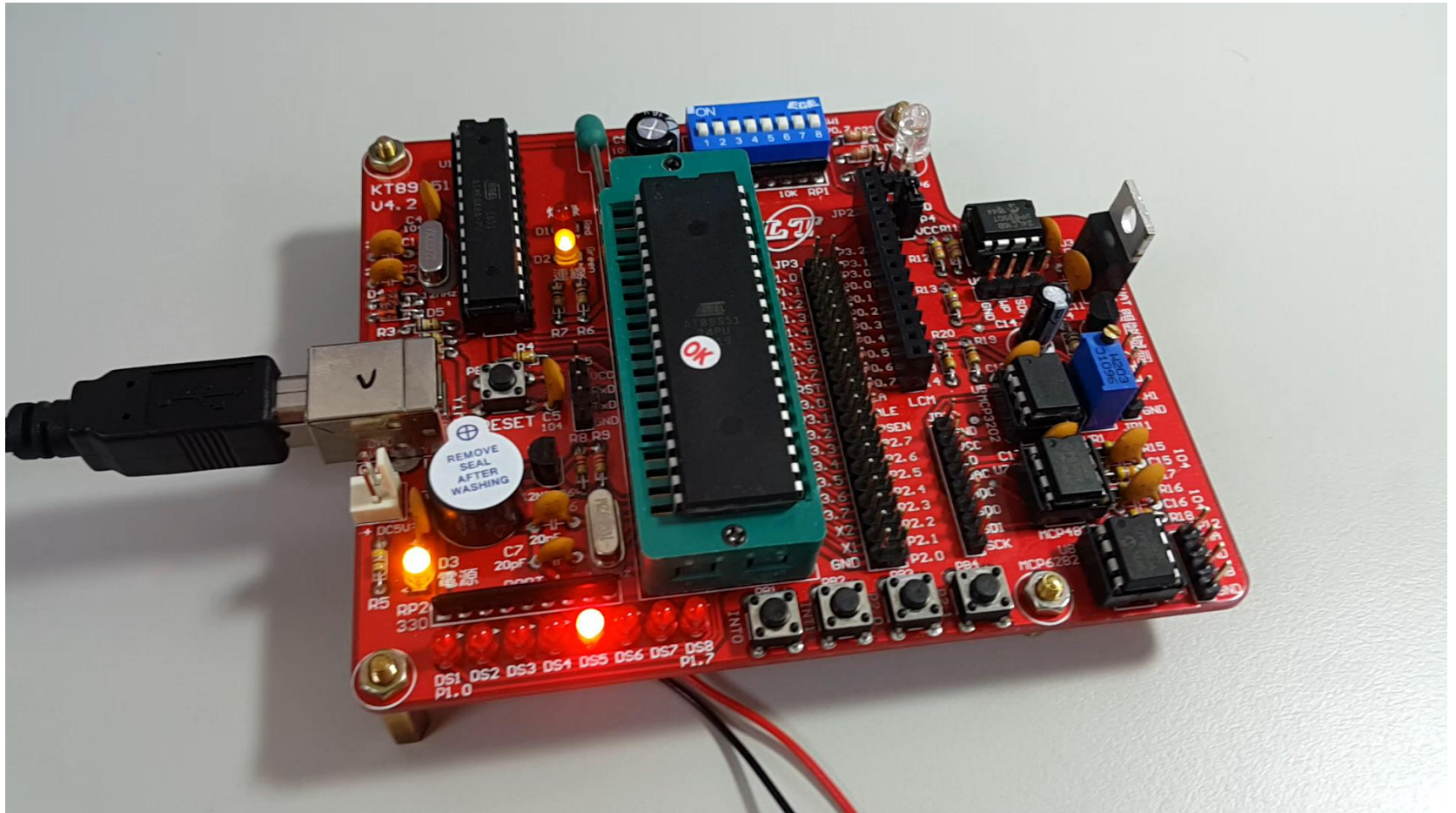
```
void EX0_int(void) __interrupt 0
{
    unsigned char j;
    j=LED;
    P1=0x0F;
    delay_ms(10000);
    LED=j;
}
```

Example (External Interrupt)

- Level-triggered

```
void EX1_int(void) __interrupt 2
{
    unsigned char j;
    j=LED;
    P1=0x00;
    delay_ms(10000);
    LED=j;
}
```

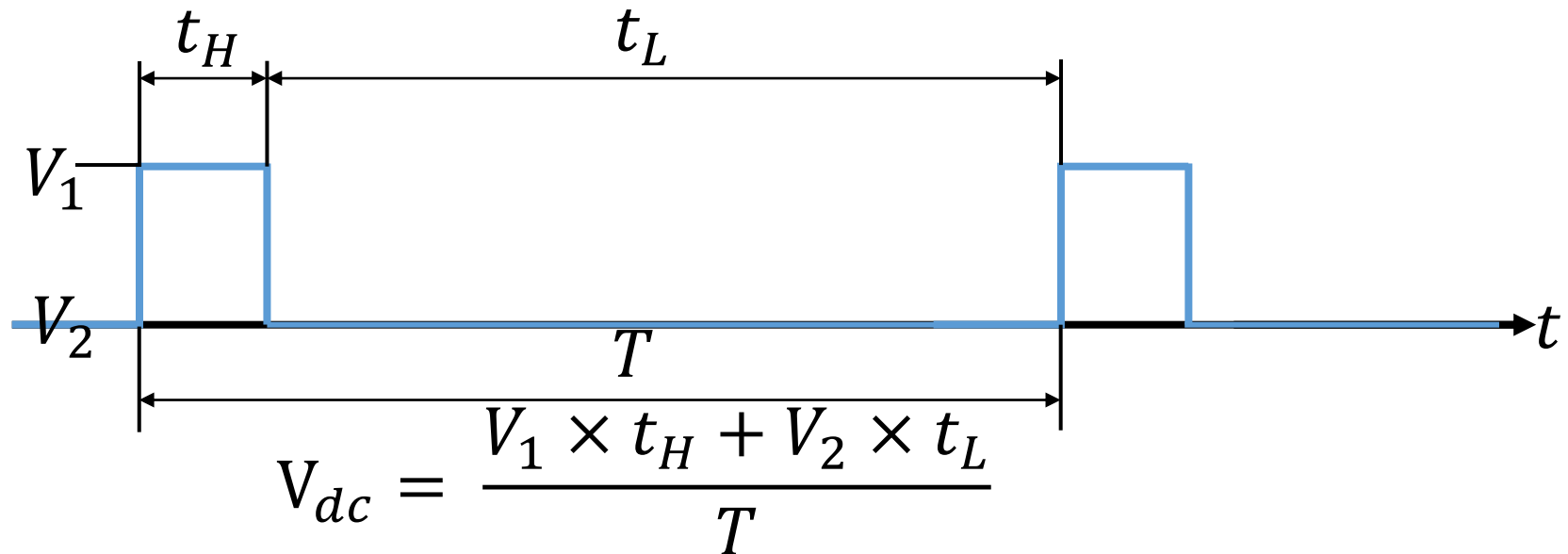
Example (External Interrupt)



PWM Introduction

- PWM is a process mainly used for getting an analog signal using a digital source.
- PWM (Pulse Width Modulation) is used in many engineering projects such as DC motor, LED dimming, etc.
- Most of the microcontrollers have special pins assigned for PWM, but 8051 microcontroller doesn't have the function.

V_{dc} and Duty Cycle

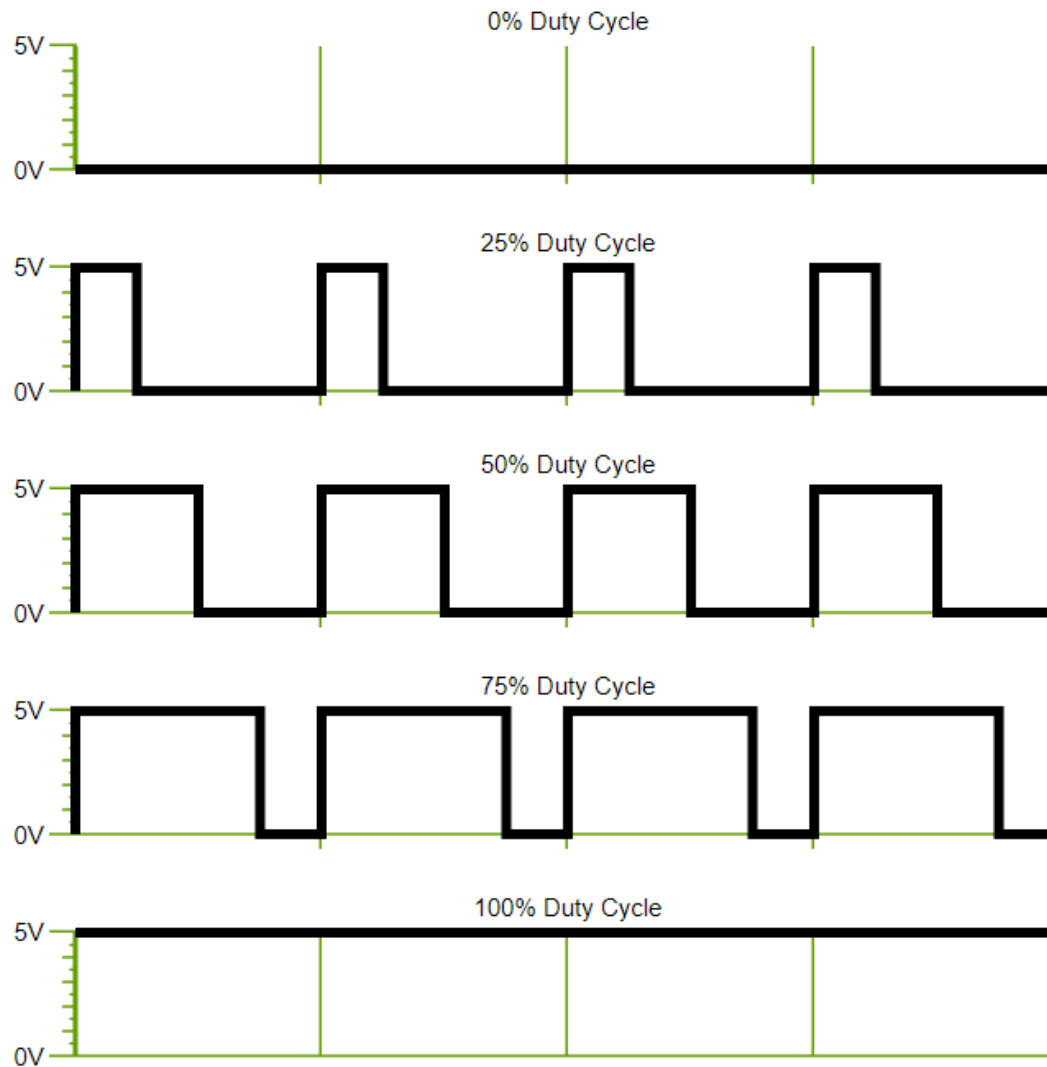


For 8051 $V_1 = V_{CC} = 5V$, $V_2 = 0V$

$$V_{dc} = V_{CC} \times \frac{t_H}{T}$$

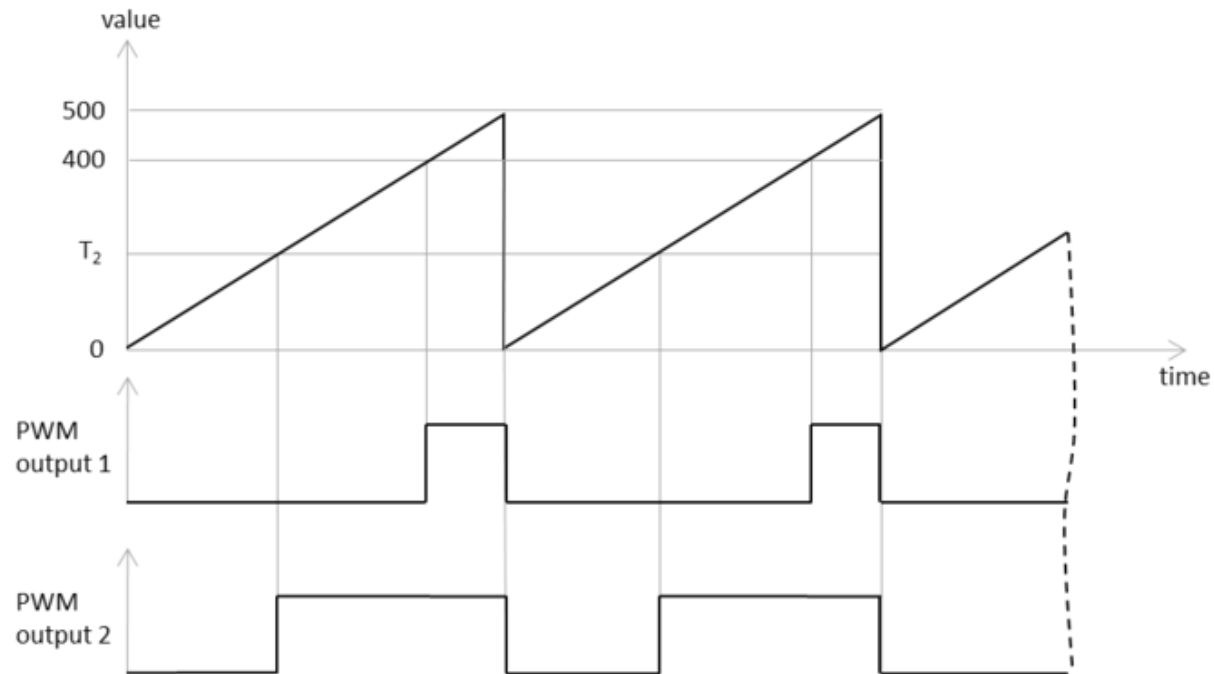
$$\frac{t_H}{T} \text{ (duty cycle)}$$

Duty Cycle



Implementing PWM on 8051

- Since 8051 doesn't have special pins assigned for PWM, the basic idea behind PWM implementation on 8051 is using timers and switching port pin high/low at defined intervals.



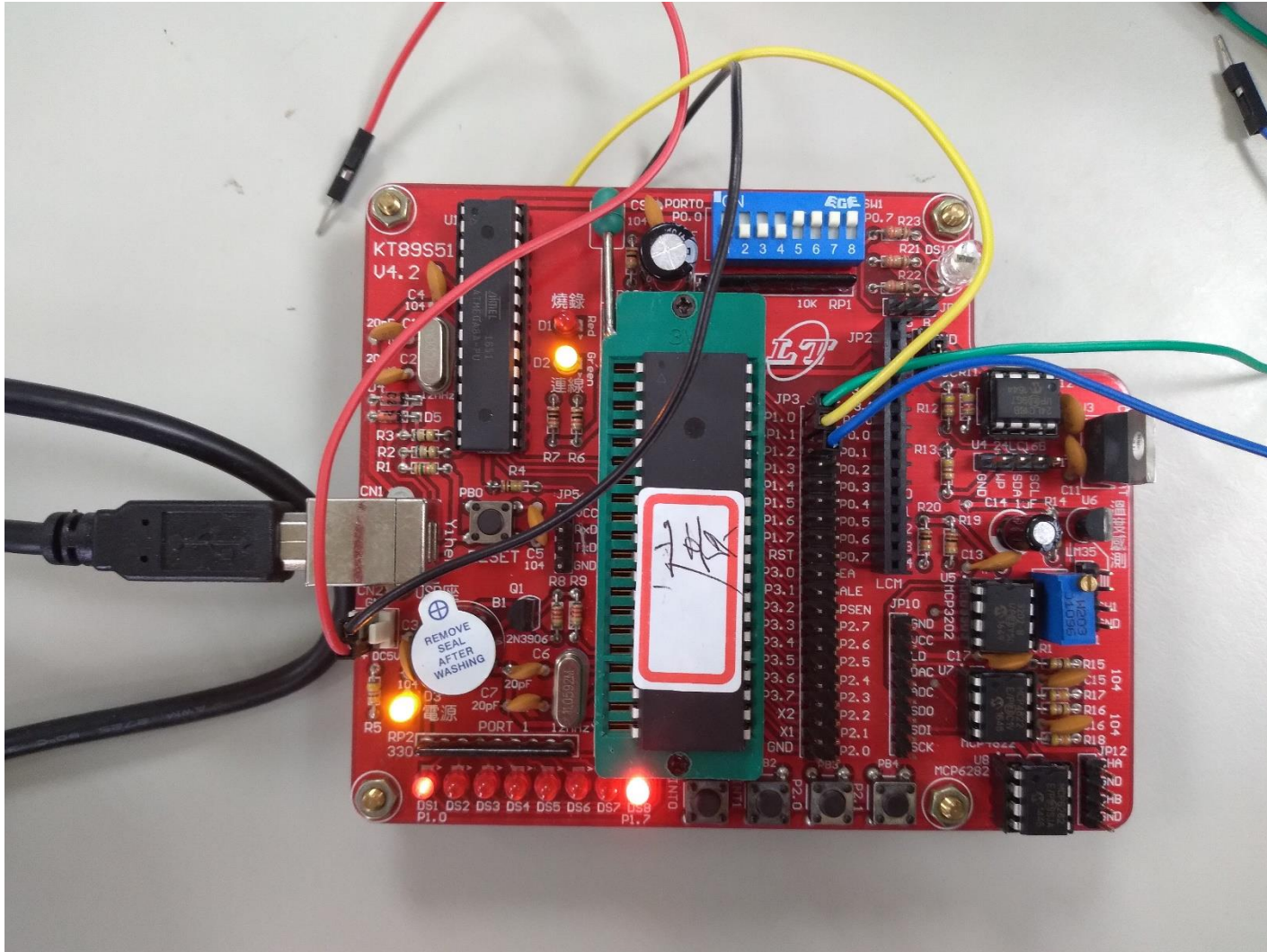
Example (LED)

```
#include "8051.h"
#include "delay.h"
#define PWM_Pin P1_0
#define cnt 100
#define freq 65536 - cnt
#define PWM_CNT_UPPER 100
int pwm_count=0;
int duty_cycle = 10;
int main(void)
{
    int i;
    P1 = 0x7F;
    TMOD = 0x01;
    TH0 = freq >> 8;
    TL0 = freq & 0xFF;
    ET0 = 1;
    EA = 1;
    TR0 = 1;
    while(1);
}
```

```
void PWM_Dimming_LED(void) __interrupt 1
{
    pwm_count++;
    if(pwm_count>=PWM_CNT_UPPER)
        pwm_count=0;

    if(duty_cycle > pwm_count)
        P1_0 = 0;
    else
        P1_0 = 1;
    TH0 = freq >> 8;
    TL0 = freq & 0xFF;
}
```

Example (LED)



Example (LED Dimming)

```
int main(void)
{
    int i;
    P1 = 0x7F;
    TMOD = 0x01;
    TH0 = freq >> 8;
    TL0 = freq & 0xFF;
    ET0 = 1;
    EA = 1;
    TR0 = 1;

    while(1)
    {
        for(i = 0; i <= PWM_CNT_UPPER; i++)
        {
            duty_cycle = i;
            delay_ms(100);
        }
    }
}
```

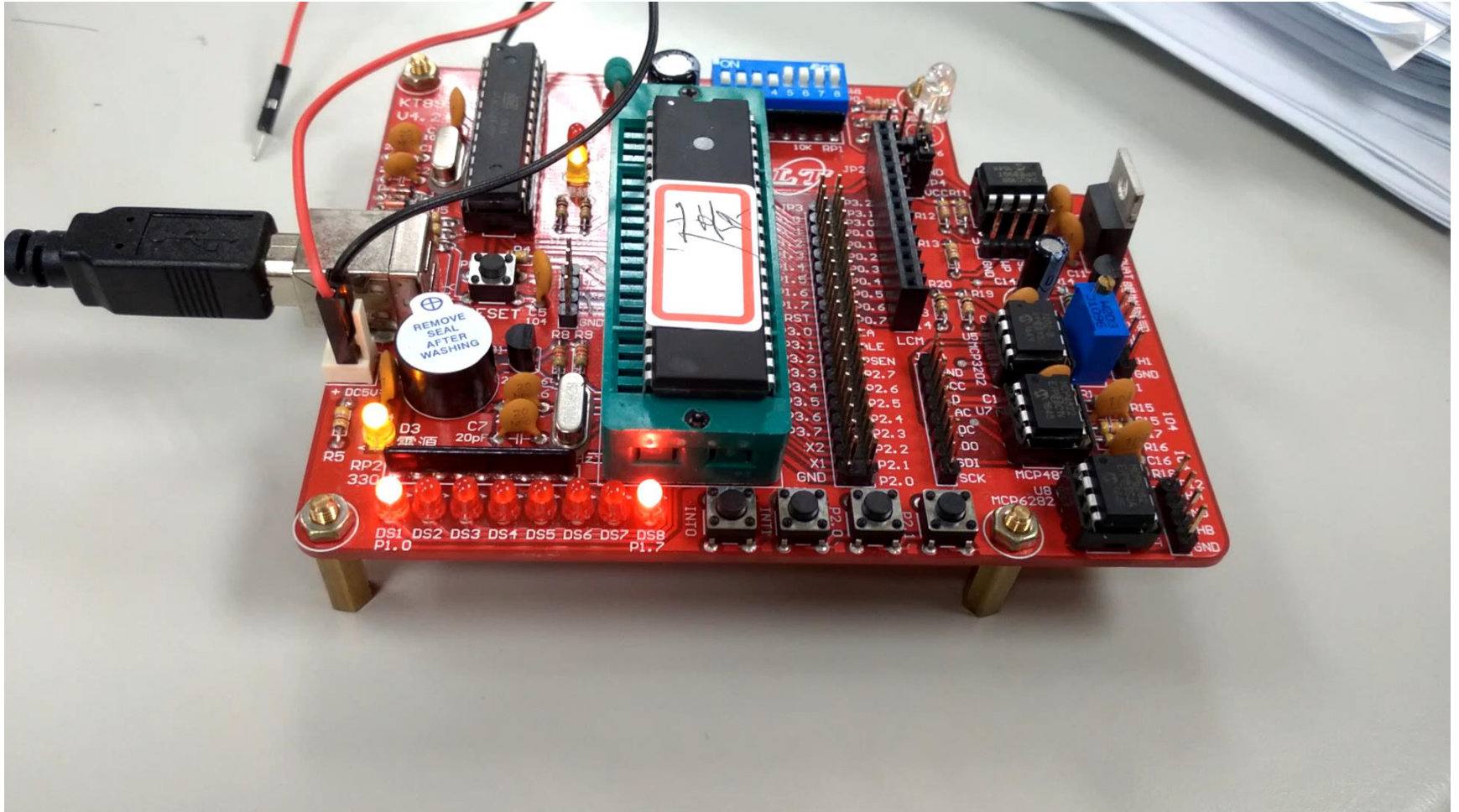
Example (LED Dimming)

```
void PWM_Dimming_LED(void) __interrupt 1
{
    TH0 = freq >> 8;
    TL0 = freq & 0xFF;

    pwm_count++;
    if(pwm_count >= PWM_CNT_UPPER)
        pwm_count = 0;

    if(duty_cycle > pwm_count)
        PWM_Pin = 0;
    else
        PWM_Pin = 1;
}
```

Example (LED Dimming)



Reference

- <https://www.electronicwings.com/8051/8051-interrupts>
- <https://www.circuitstoday.com/external-interrupts-handling-in-8051>
- <https://www.engineersgarage.com/8051-microcontroller/dc-motor-speed-control-using-8051-microcontroller/>